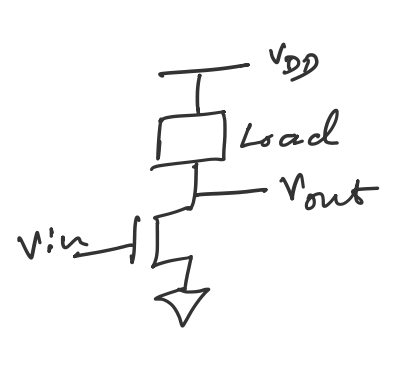
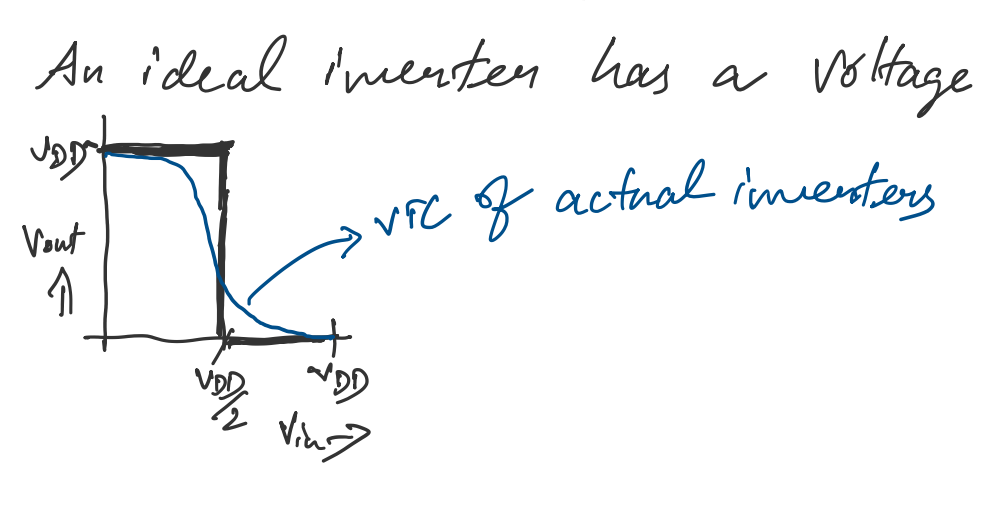
Inverter DC characteristics

Friday, August 8, 2025 9:56 PM

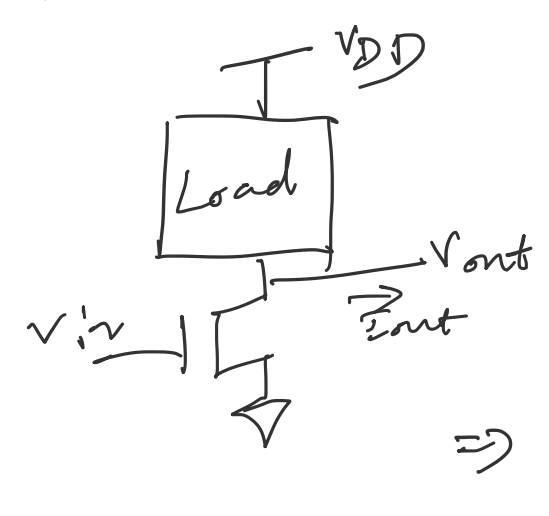


Typically V_{out} drives the gate terminal of the subsequent stage. This implies there is no current drawn out of V_{out} in steady state.

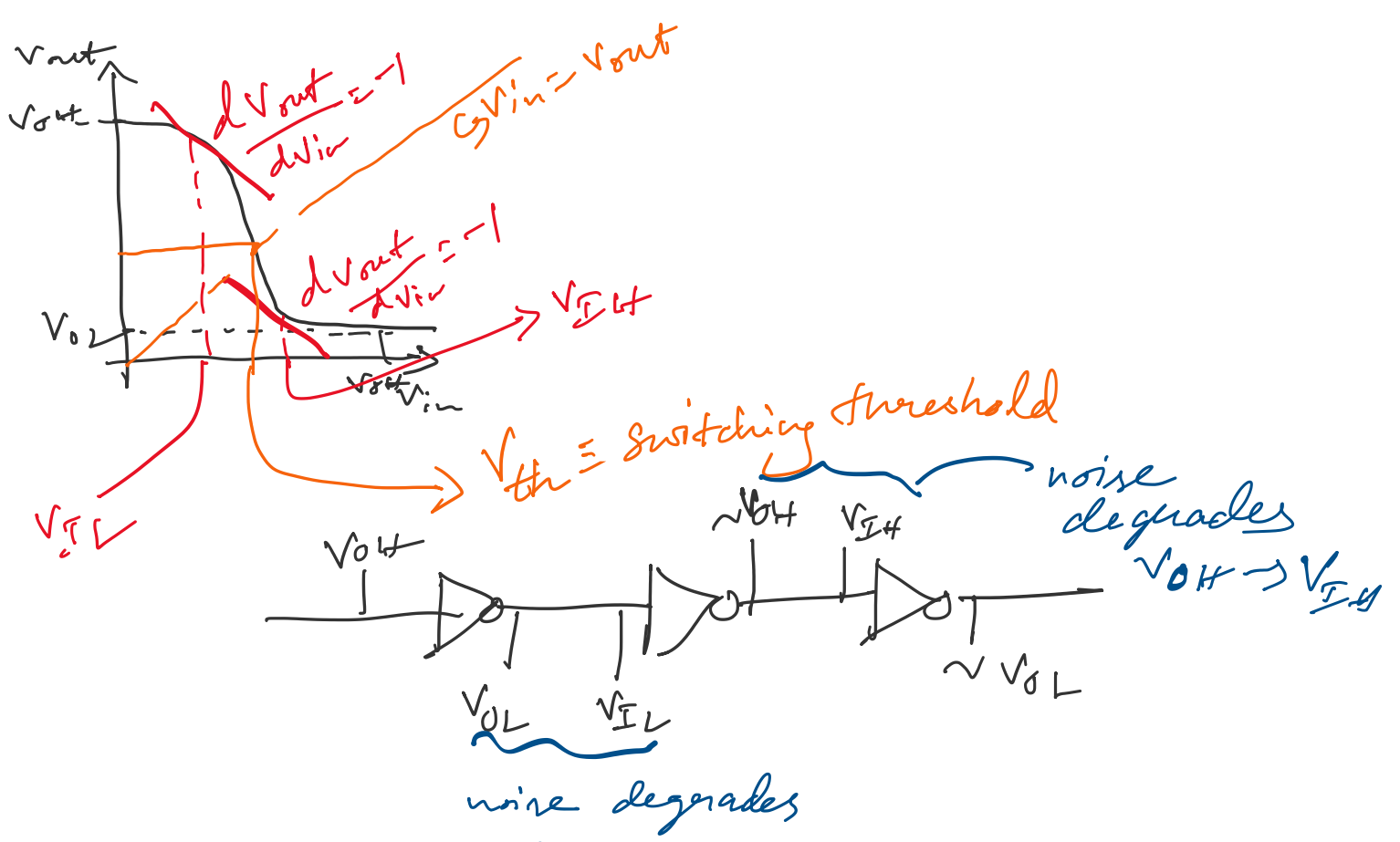
An ideal inverter has a voltage transfer characteristics (VTC)



Consider a nMOS inverter using a generic load:



typically V_{out} drives another gate input of the subsequent nMOS $\Rightarrow I_{out} \approx 0$ at steadystate



Five quantities of interest in VTC

V_{IL} = maximum input voltage that is considered logic "0"

V_{IH} = " for logic "1"

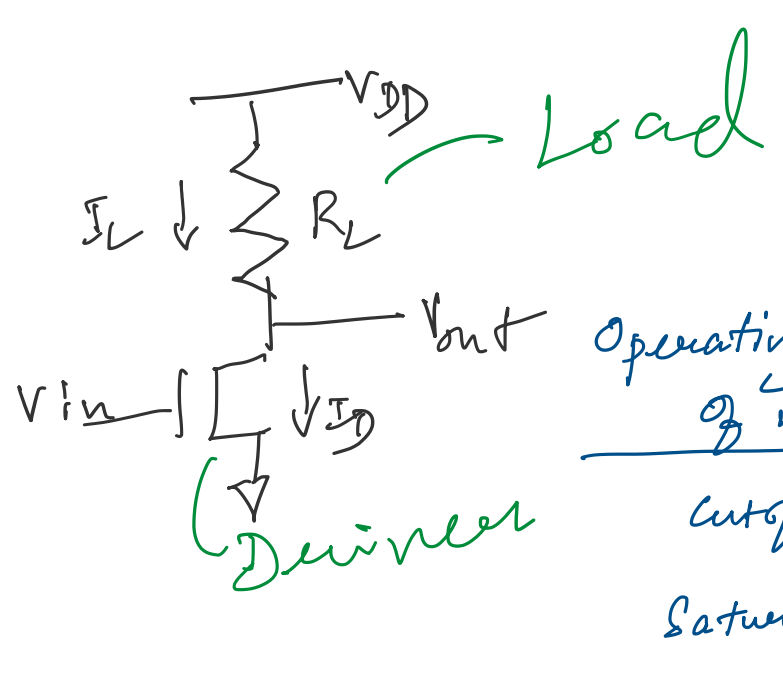
V_{OH} = output High

V_{OL} = output Low

V_{th} = switching threshold ($V_{in} = V_{out}$)

Since the PDN is made of nMOS $V_{OH} = V_{DD}$ for the gates we will be seeing.

Resistive load



Operating mode of nMOS	V_{in}
cutoff	$V_{in} \le V_{Tn}$
Saturation	$V_{Tn} < V_{in} \le V_{out} + V_{Tn}$
Linear	$V_{in} > V_{out} + V_{Tn}$

$$\boxed{V_{OH} = V_{DD}}$$

V_{OL} Calculation

$V_{out} = V_{OL}$, $V_{in} = V_{OH} = V_{DD}$

typically a small value

$\Rightarrow V_{in} > V_{out} + V_{Tn} \Rightarrow$ nMOS is in linear region

Ignoring channel length modulation:

KCL

Assuming $V_{out} = V_{OL} << E_cL$

$$I_L = \frac{V_{DD} - V_{out}}{R_L} = I_D = \frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)_n \left[2(V_{in} - V_{Tn})V_{out} - V_{out}^2\right] \times \frac{\cancel{E_c L}}{\cancel{E_c L + V_{out}}}$$

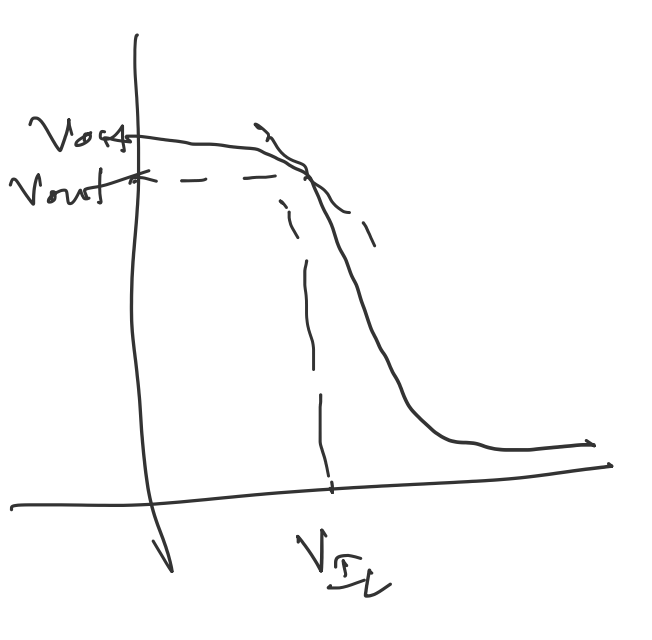
$$\frac{V_{DD} - V_{OL}}{R_L} = \frac{k_n}{2}\left[2(V_{DD} - V_{Tn})V_{OL} - V_{OL}^2\right] \quad \text{where } k_n = \mu_n C_{ox}\left(\frac{W}{L}\right)_n$$

$$V_{OL} = V_{DD} - V_{Tn} + \frac{1}{k_n R_L} - \sqrt{\left(V_{DD} - V_{Tn} + \frac{1}{k_n R_L}\right)^2 - \frac{2V_{DD}}{k_n R_L}}$$

HW Calculate R_L for $V_{OL} = 100\,mV$ assuming $V_{DD} = 1.8\,V$, $C_{oxn} = 0.008834\,F/m^2$, $\mu_n = 0.025\,m^2/Vs$, $v_{sat,n} = 8 \times 10^4\,m/s$, $\left(\frac{W}{L}\right) = \frac{0.42\,\mu m}{0.15\,\mu m}$

V_{IL} Calculation

when $V_{in} = V_{IL}$, $\frac{dV_{out}}{dV_{in}} = -1$



$V_{out} = V_{DS} \approx V_{OH}$ large value

$V_{in} = V_{GS}$ small value

$\Rightarrow$ nMOS is in saturation

$$\frac{k_n}{2}\frac{(V_{in} - V_{Tn})^2 \times \cancel{E_cL}}{\cancel{V_{in} - V_{Tn} + E_cL}} = \frac{V_{DD} - V_{out}}{R_L} \quad \text{(KCL)}$$

Assuming $V_{in} - V_{Tn} << E_cL$

$$\frac{k_n}{2}(V_{in} - V_{Tn})^2 \approx \frac{V_{DD} - V_{out}}{R_L} \quad \text{—①}$$

Taking derivative of ① w.r.t V_{in}

$$k_n(V_{in} - V_{Tn}) = -\frac{1}{R_L}\frac{dV_{out}}{dV_{in}}$$

Using $\frac{dV_{out}}{dV_{in}} = -1$, $V_{in} = V_{IL}$

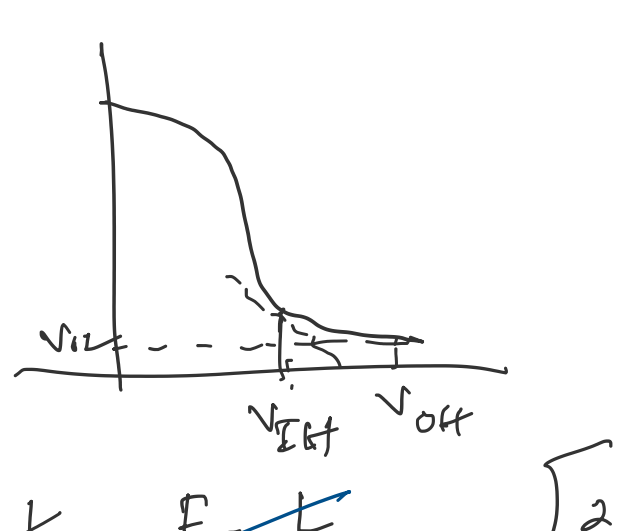
$$\boxed{V_{IL} = V_{Tn} + \frac{1}{k_n R_L}}$$

HW Calculate V_{IL} for $V_{OL} = 100\,mV$.

V_{IH} Calculation

$V_{in} = V_{IH}$, $V_{out} \approx V_{OL}$

Driver = linear region



Assuming $V_{out} << E_cL$

$$\text{KCL} \quad \frac{V_{DD} - V_{out}}{R_L} = \frac{k_n}{2}\frac{\cancel{E_cL}}{\cancel{E_cL + V_{out}}}\left[2(V_{in} - V_{Tn})V_{out} - V_{out}^2\right] \quad \text{—①}$$

Derivative w.r.t V_{in}

$$-\frac{1}{R_L}\frac{dV_{out}}{dV_{in}} = \frac{k_n}{2}\left(2V_{out} + 2(V_{in} - V_{Tn})\frac{dV_{out}}{dV_{in}} - 2V_{out}\frac{dV_{out}}{dV_{in}}\right)$$

Using $\frac{dV_{out}}{dV_{in}} = -1$ for $V_{in} = V_{IH}$

$$V_{IH} = V_{Tn} - \frac{1}{k_n R_L} + 2V_{out} \quad \text{—②}$$

Solve ① and ② to get

$$V_{IH} = V_{Tn} + \sqrt{\frac{8V_{DD}}{3k_n R_L}} - \frac{1}{k_n R_L}$$

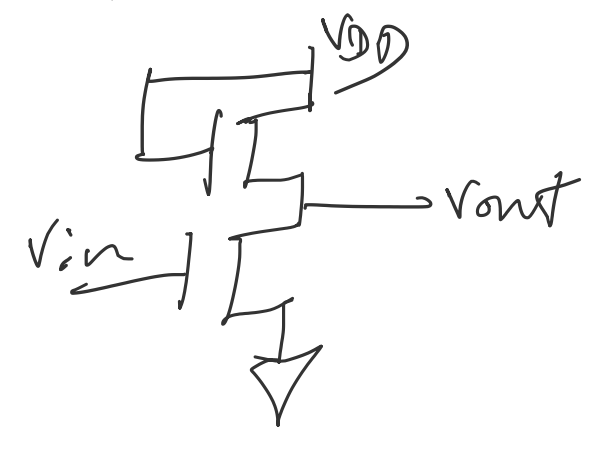
Issues with using resistive load inverter

① Poly or diffusion segments are used to create resistors. For small V_{OL} long poly/diff. segments are required which translates to a large layout area.

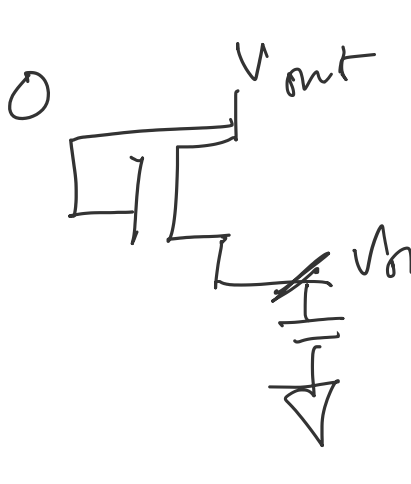
② For $V_{in} = V_{OH}$, $V_{out} = V_{OL} > 0 \Rightarrow$ there is a DC current from V_{DD} to GND through the R_L & nMOS $\Rightarrow$ static power dissipation.

Pseudo nMOS

We can replace R_L with any active load. Simplest load is to use another nMOS.

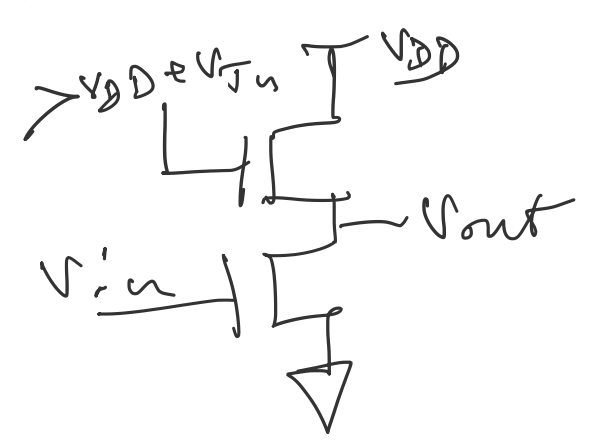


For $V_{in} = 0$



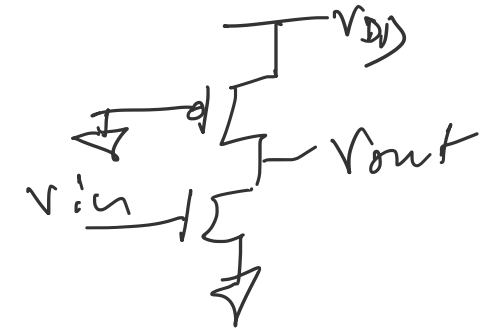
V_{out} can only rise till $V_{DD} - V_{Tn}$ before the top nMOS turns OFF.

One way to fix this is to connect the gate of top nMOS to another supply.



Dual supply voltages for each gate is problematic.

A pMOS can replace the load nMOS



$\Rightarrow V_{OH} = V_{DD}$

For load pMOS $V_{SG} = V_{DD}$, $V_{SD} = V_{DD} - V_{out}$

for saturation $V_{SD} \ge V_{SG} - |V_{Tp}|$

$V_{DD} - V_{out} \ge V_{DD} - |V_{Tp}|$

$V_{out} \le |V_{Tp}|$

As V_{out} increases pMOS goes from sat. to linear.

For V_{OL} calculation, nMOS is linear & pMOS is in sat.

$$V_{OL} = V_{DD} - V_{Tn} - \sqrt{(V_{DD} - V_{Tn})^2 - \frac{k_p}{k_n} \times \frac{E_{cp}L(V_{DD} - |V_{Tp}|)^2}{E_{cp}L + V_{DD} - |V_{Tp}|}}$$

For V_{IL}, nMOS is in sat. & pMOS is linear

$$V_{IL} = V_{Tn} + \frac{k_p}{k_n}(V_{out} - |V_{Tp}|) \quad \text{—①}$$

use ① & KCL to get V_{IL}

For V_{IH}, nMOS is linear & pMOS is sat.

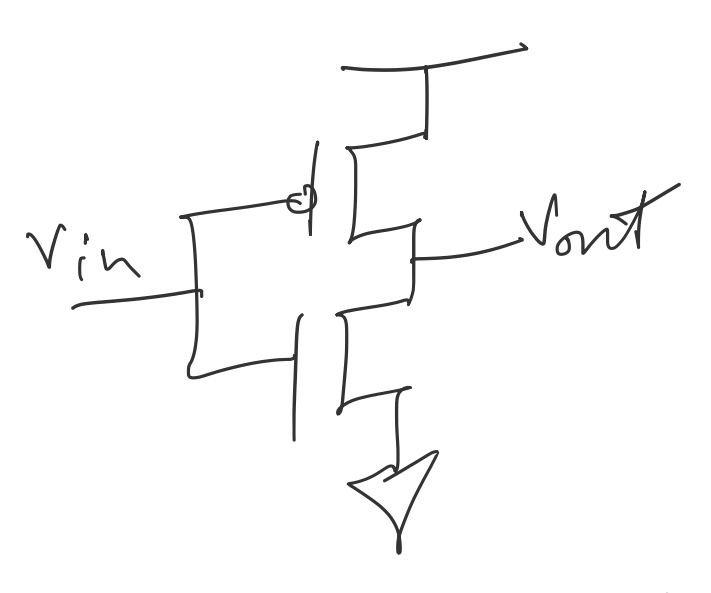
$$V_{IH} = V_{Tn} + \frac{2}{\sqrt{3}}\sqrt{\frac{k_p}{k_n}}(V_{DD} - |V_{Tp}|)$$

All the quantities V_{OL}, V_{IL}, V_{IH} are a function of $\boxed{\frac{k_n}{k_p}}$.

$\Rightarrow$ pseudo nMOS = ratioed logic

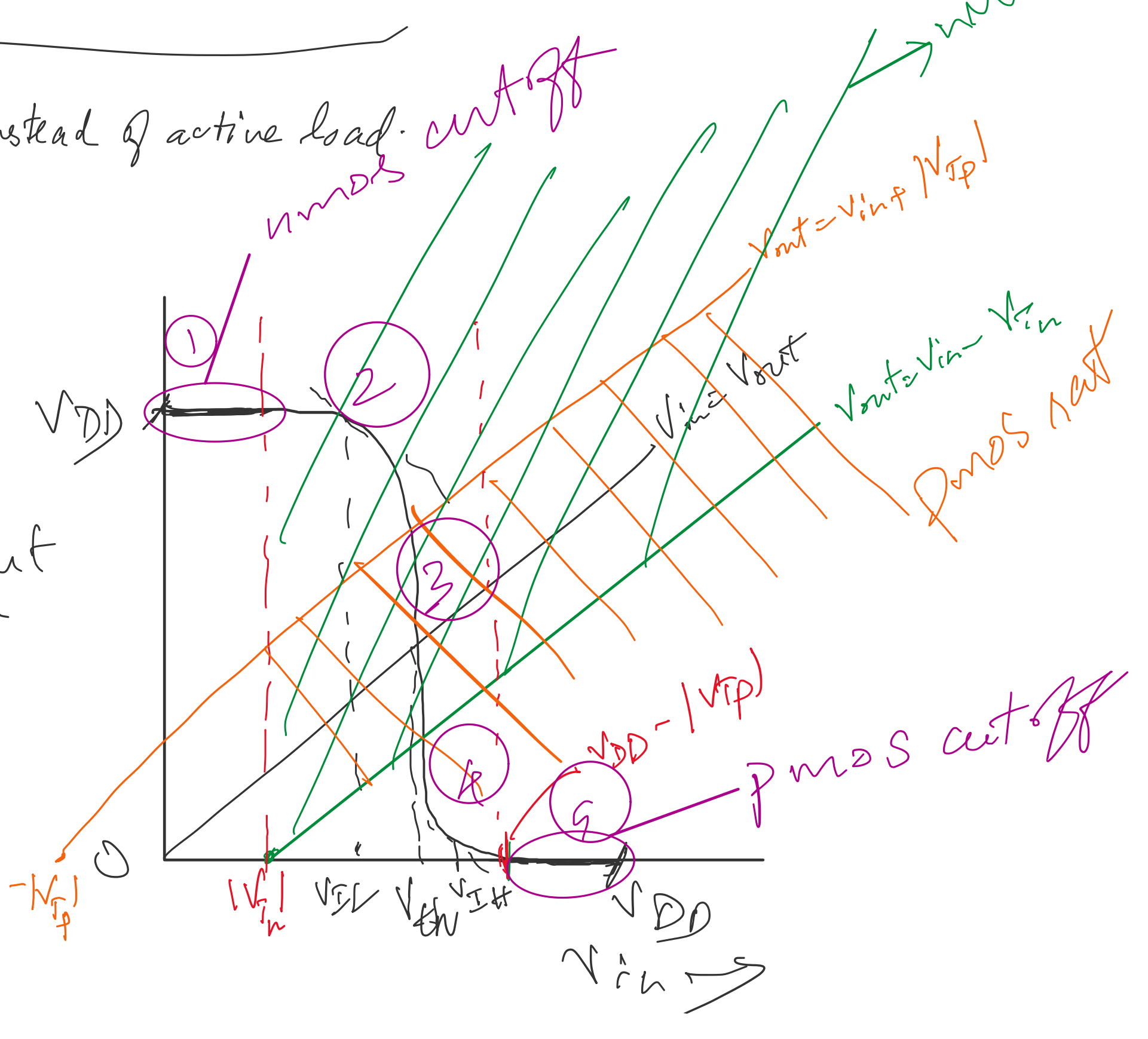
CMOS

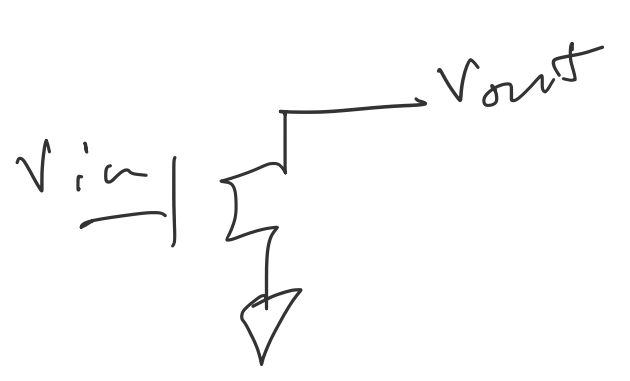
pMOS can be used for logic high instead of active load.



$V_{OH} = V_{DD}$

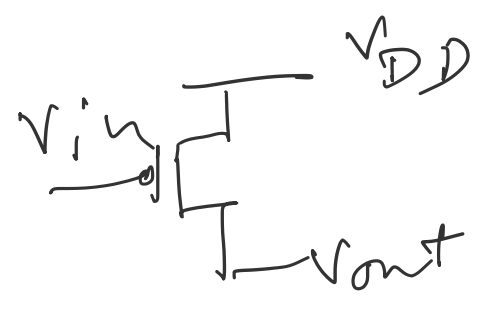
$V_{OL} = 0$





for saturation

$V_{out} \ge V_{in} - V_{Tn}$



$$\underbrace{V_{DD} - V_{out}}_{V_{SD}} \ge \underbrace{V_{DD} - V_{in}}_{V_{SG}} - |V_{Tp}|$$

$V_{out} \le V_{in} + |V_{Tp}|$

	nmos	pmos
①	cutoff	linear
②	sat	linear
③	sat	sat
④	linear	sat
⑤	linear	cutoff

Switching threshold

$V_{th} = V_{in} = V_{out}$ } Both n/p mos are in sat.

$$\frac{k_n}{2}\frac{(V_{GSn} - V_{Tn})^2 E_{cn}L_n}{E_{cn}L_n + V_{GSn} - V_{Tn}} = \frac{k_p}{2}\frac{(V_{SGp} - |V_{Tp}|)^2 E_{cp}L_p}{E_{cp}L_p + V_{SGp} - |V_{Tp}|}$$

For symmetric VTC $V_{in} = V_{out} = \frac{V_{DD}}{2} = V_{th}$

$$\frac{\mu_n C_{ox}}{2}\left(\frac{W}{L}\right)_n \frac{\left(\frac{V_{DD}}{2} - V_{Tn}\right)^2 \times \cancel{E_{cn}L_n}}{\cancel{E_{cn}L_n + \frac{V_{DD}}{2}} - V_{Tn}} = \frac{\mu_p C_{ox}}{2}\left(\frac{W}{L}\right)_p \frac{\left(V_{DD}/2 - |V_{Tp}|\right)^2 \cancel{E_{cp}L_p}}{\cancel{E_{cp}L_p + \frac{V_{DD}}{2} - |V_{Tp}|}}$$

$\frac{V_{DD}}{2} - V_{Tn} << E_{cn}L_n$

If PDK provides $V_{Tn} = |V_{Tp}|$

$$\left(\frac{W}{L}\right)_n \mu_n = \left(\frac{W}{L}\right)_p \mu_p$$

$L_n = L_p = L_{min}$

$$\Rightarrow \frac{W_p}{W_n} = \frac{\mu_n}{\mu_p}. \quad 2 < \frac{\mu_n}{\mu_p} < 3$$

we want W_p to be an integer multiple of W_n

$W_p = 2W_n$ for the rest of the course.

Sky130 params

$W_{nmin} = 0.42\,\mu m$

$L_n = L_p = 0.15\,\mu m$

$E_{cn}L_n = \frac{2 v_{satn}}{\mu_n} L_n \approx 0.96V$

$E_{cp}L_p = \frac{2 v_{satp}}{\mu_p} L_p \approx 0.83V$

$V_{DD} = 1.8\,V$

$V_{Tn} = |V_{Tp}| = 0.7V$

HW Calculate V_{IL}, V_{IH}, NM_H, NM_L for CMOS with $\frac{W_p}{W_n} = 2$

For V_{IL} nMOS = SAT, pMOS = linear

① $\frac{dV_{out}}{dV_{in}} = -1$, $V_{in} = V_{IL}$

② KCL $I_{Dn} = I_{Dp}$